Week 8, Lecture 1: Priors and Posteriors

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** On Wednesday we built Bayes' rule for parameter values and worked out why a Beta prior and a binomial likelihood give a Beta posterior. Today we use that result on a real example with two different priors, learn how to summarize a posterior, and talk about common misunderstandings and why Bayesian methods are so widely used. You'll also get a short take-home WOD on Newton's method, due Wednesday, to get ready for Wednesday's lecture on optimization.
- **Read:** Bolker 6.2.2 (printed p. 237-243) and 6.4.2 (p. 258-260), if you haven't yet. For the take-home and Wednesday: Bolker 7.1-7.2 (p. 294-310).
- **Review:** your notes on Beta-binomial conjugacy from Wednesday, the Beta distribution (Bolker 4.5.2.5, p. 176), and the log-odds, $\log(p/(1-p))$, from Wednesday's logistic regression opening.

Overview (instructor only)

This file has two parts, both instructor-only. **Part 1** is read beforehand. **Part 2** (after the page break, large print) is the podium copy.

Beat	Time
WOD: "How much did the prior matter?"	~8 min
Lecture	~32 min
Discussion	~8 min

Where this came from. Sections 2-5 and the discussion were planned for Week 7 Wednesday, which instead covered logistic regression, Bayes' rule for parameters, and the conjugacy math (see `NOTES_Week7_Lecture2_BayesianInference.Rmd`). Start with the germination example.

Pacing. Both Week 7 lectures covered about half of what was planned. Core: sections 1-2, confusions (a), (e) from section 4 (with a quick callback for (b)), two reasons from section 5, and section 6. Section 3 and the rest of 4-5 are "if time." The optimization preview planned earlier is gone: Wednesday covers it, and the take-home WOD (below) does the setup.

WODs.

- **In class:** `wod/WOD_Week8_Lecture1_HowMuchDidThePriorMatter.Rmd` (key in `wod/answer_keys/`). Three analysts fit the same tortoise survival data with different priors; each panel shows prior and posterior only. Students reconstruct the shared likelihood (best read from Analyst A's weak-prior posterior), rank how much each result depends on its prior, and tell a manager which number to use (posterior means 0.68, 0.84, 0.62). It works as a prior sensitivity analysis in miniature. The trap is Analyst B: narrowest posterior, but that's the prior, not more data. Debrief straight into section 2: the germination supplier prior is worth 200 seeds against 40 of data, heavier relative to the data than any analyst's prior.
- **Take-home, due Wednesday:** `wod/WOD_Week8_Lecture1_NewtonByHand.Rmd` (key in `wod/answer_keys/`). Pika NLL derivatives, finite differences, and Newton steps by hand, ending on an overshoot from $\lambda = 7$. Hand it out at the end of class with a one-line pitch: "Wednesday we open up `optim()`; this is the first piece."
- `wod/WOD_Week7_Lecture2_CurvatureByHand.Rmd` stays shelved; the take-home absorbs its pika part.

Source material: Bolker 4.3.3 (priors, and why flat priors aren't neutral), 6.2.2.1 (binomial/Beta conjugate example; the numbers below are his, recast as a germination trial), 6.4.2 (credible intervals); legacy `LECTURE6.Rmd` ("Bayesian Analysis #1: Concepts": mean vs. mode, credible intervals).

Example choice. The conjugate example is a seed-germination trial rather than Bolker's tadpoles. The numbers (30 of 40, and a prior worth 120 of 200) are Bolker's from p. 237-238, so students who did the reading can check them, but the setting is one the agricultural and restoration students will recognize.

Where this sits in the week

- **Last week:** likelihood intervals (Monday), the quadratic approximation and profiles (Lab 6), then logistic regression and the start of Bayesian inference (Wednesday).
- **Today:** finish the one-parameter Bayesian material. Hand out the Newton's method take-home.
- **Tuesday:** group meetings on final project proposals (no lab).
- **Wednesday:** optimization (Bolker 7.1-7.2), opening with the take-home.
- **Coming up:** MCMC and Lab 7 on Bayesian inference (Week 9).

Part 1: Lecture content

1. Big picture and recap (~3 min)

Loop on the board:

$$\text{THEORY} \rightarrow \text{MODEL(S)} \rightarrow \text{DATA (fit)} \rightarrow \text{COMPARE} \rightarrow \text{repeat}$$

- Same model, same likelihood; Bayesian inference describes our uncertainty about a parameter with a probability distribution.
- Recap: posterior $\propto$ likelihood $\times$ prior. With a Beta prior and binomial data, the posterior is Beta($\alpha + k$, $\beta + n - k$).
- Today's question: how much does the prior matter, and what do we report?

2. A worked example: germination with two priors (~10 min)

Bridge from the WOD. The tortoise analysts showed by eye that the same data can give different answers under different priors. Here we put numbers on it: how much is a prior "worth," in data?

Setting. A restoration crew tests a native seed lot: 30 of 40 seeds germinate. What is the lot's germination probability p?

The likelihood is binomial: $\mathcal{L}(p) \propto p^{30}(1-p)^{10}$.

Recap from Wednesday. With a Beta(α, β) prior, the posterior is Beta($\alpha + k$, $\beta + n - k$): just add counts.

- Interpretation (Bolker p. 237): $\alpha - 1$ and $\beta - 1$ act like earlier successes and failures. Beta(1, 1) is flat: no prior seeds.
- **Flat prior:** posterior Beta(31, 11).
- **Informative prior:** the supplier's records say 120 of 200 seeds from earlier lots germinated. Prior Beta(121, 81); posterior Beta(151, 91).

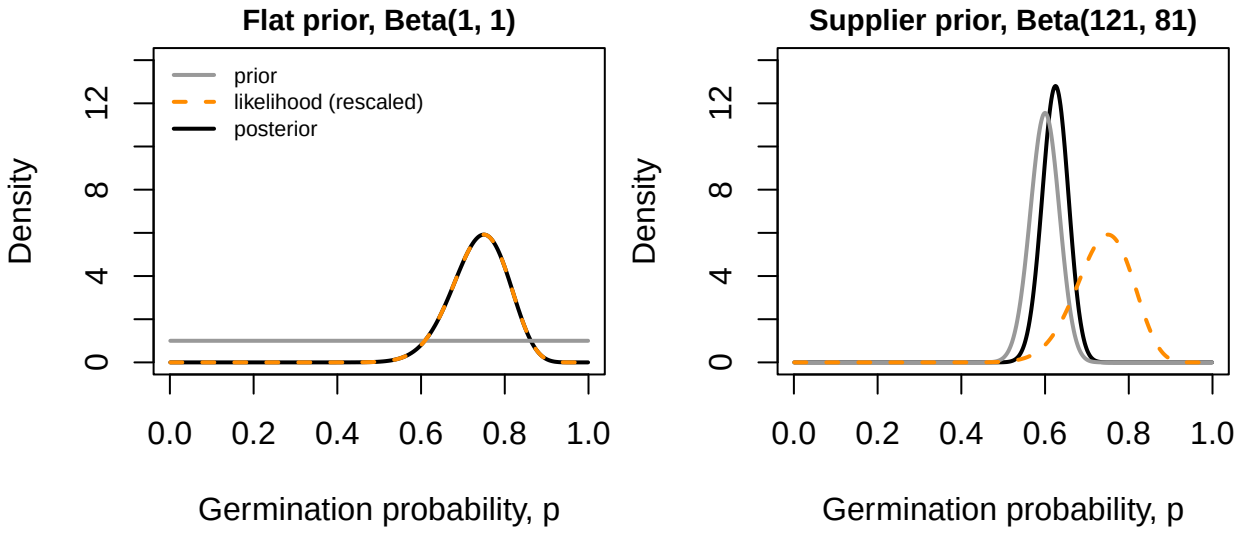
```
k <- 30; n <- 40
post_flat <- c(1 + k, 1 + n - k)
post_info <- c(121 + k, 81 + n - k)
beta_mean <- function(ab) ab[1] / sum(ab)
beta_mode <- function(ab) (ab[1] - 1) / (sum(ab) - 2)
summ <- rbind(
  flat = c(mean = beta_mean(post_flat), mode = beta_mode(post_flat),
           qbeta(c(0.025, 0.975), post_flat[1], post_flat[2])),
  informative = c(mean = beta_mean(post_info), mode = beta_mode(post_info),
                  qbeta(c(0.025, 0.975), post_info[1], post_info[2]))
)
colnames(summ)[3:4] <- c("lower95", "upper95")
round(summ, 3)
```

```
##              mean  mode lower95 upper95
## flat        0.738 0.750   0.597   0.858
## informative 0.624 0.625   0.562   0.684
```

```
f2 <- function(v) formatC(v, format = "f", digits = 2)
```

```
p_seq <- seq(0.001, 0.999, length.out = 500)
lik_scaled <- dbeta(p_seq, k + 1, n - k + 1)   # likelihood rescaled to area 1
```

```r
plot_panel <- function(prior_ab, post_ab, title) {
  plot(p_seq, dbeta(p_seq, post_ab[1], post_ab[2]), type = "l", lwd = 2,
       ylim = c(0, 14), xlab = "Germination probability, p", ylab = "Density",
       main = title, cex.main = 0.9)
  lines(p_seq, dbeta(p_seq, prior_ab[1], prior_ab[2]), lwd = 2, col = "grey60")
  lines(p_seq, lik_scaled, lwd = 2, lty = 2, col = "darkorange")
}
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
plot_panel(c(1, 1), post_flat, "Flat prior, Beta(1, 1)")
legend("topleft", c("prior", "likelihood (rescaled)", "posterior"),
       lwd = 2, lty = c(1, 2, 1), col = c("grey60", "darkorange", "black"),
       bty = "n", cex = 0.7)
plot_panel(c(121, 81), post_info, "Supplier prior, Beta(121, 81)")
```



- **Flat prior:** the posterior *is* the rescaled likelihood. Its mode is 0.75, the MLE; its mean is 0.738, slightly pulled toward 0.5.
- **Supplier prior:** the posterior sits between prior and data, closer to the prior (mean 0.624), because the prior is "worth" 200 seeds and the data only 40.
- **More data wins.** If the crew had tested 400 seeds with 300 germinating, the same supplier prior gives a posterior mean of 0.699: the data now outweigh the prior two to one.
- **Updating in sequence.** Today's posterior is tomorrow's prior. Testing 40 seeds and then 40 more gives the same answer as testing 80 at once (Bolker p. 237).

3. Summarizing a posterior (~4 min)

- **Point estimates.** Posterior mean (most common), median, or mode. With a flat prior, the mode equals the MLE. For skewed posteriors the three differ; Bolker prefers the mean because it uses the whole distribution.
- **Credible interval.** The central 95% of the posterior (2.5% and 97.5% quantiles), or

the highest posterior density (HPD) interval, the shortest interval holding 95% (Bolker p. 258-259). For symmetric posteriors they coincide.

- **Interpretation.** "Given the model, the prior, and the data, there is a 95% probability that p is between 0.60 and 0.86." That is the sentence most people *want* to say about a confidence interval (Week 7 Monday's discussion question 1), and a Bayesian is allowed to say it.

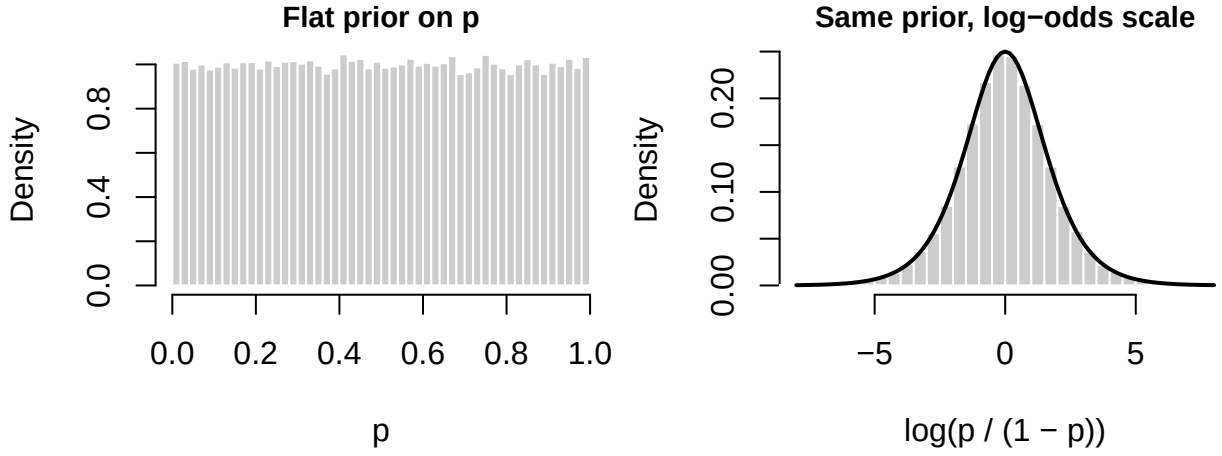
4. Where new Bayesians get stuck (~8 min)

Core: (a), (e), and a 30-second callback for (b), which was covered well in Week 7 Wednesday's lecture. The rest if time; each takes a minute or two.

(a) "The likelihood is the posterior." Only after multiplying by a prior and rescaling to area 1. The likelihood alone doesn't integrate to 1 (Week 5 WOD) and isn't a distribution for p.

(b) "A flat prior means no information." (Callback only; covered Wed., Oct. 7.) Flat on one scale isn't flat on another (Bolker 4.3.3, the bird-mass example). A uniform prior on p implies a prior on the log-odds, $\log \frac{p}{1-p}$, that is peaked at 0:

```
set.seed(746)
p_draws <- runif(1e5)
par(mfrow = c(1, 2), mar = c(4, 4, 1.5, 0.5))
hist(p_draws, breaks = 40, freq = FALSE, col = "grey80", border = "white",
     main = "Flat prior on p", xlab = "p", cex.main = 0.9)
hist(qlogis(p_draws), breaks = 60, freq = FALSE, col = "grey80", border = "white",
     main = "Same prior, log-odds scale", xlab = "log(p / (1 - p))",
     xlim = c(-8, 8), cex.main = 0.9)
curve(dlogis(x), add = TRUE, lwd = 2)
```



So "uninformative" always means "uninformative on some scale." With plenty of data it rarely matters; with little data it can.

(c) "Priors let you get any answer you want." In principle yes (Bolker p. 150: a prior

of 0 on a hypothesis means no data can ever support it). In practice: priors must be reported, reviewers can object, and the standard move is a **prior sensitivity analysis**: rerun with a few reasonable priors and see whether the conclusion changes. If it does, the data are weak and that's worth knowing.

(d) "Bayesian means MCMC." MCMC is a computing method for approximating posteriors when the denominator can't be computed. The germination example needed none. Conversely, MCMC can be used for non-Bayesian problems too.

(e) "The credible interval and the confidence interval are the same thing." With a flat prior and plenty of data the numbers are often close (Week 7 pikas: likelihood ratio [1.82, 4.61]; with a flat prior the posterior is Gamma(19, 6), credible interval [1.91, 4.74]), but the statements differ: a credible interval is a probability statement about the parameter; a confidence interval is a statement about the procedure.

(f) "A high posterior probability proves the model." Every posterior is conditional on the model being right. A Bayesian model still has to be confronted with data (does it predict data that look like what we saw?) and compared with rivals. The loop on the board doesn't change.

5. Why Bayesian methods are so widely used (~5 min)

Interpretation.

- Direct probability statements about parameters and hypotheses: "95% probability the germination rate is between 0.60 and 0.86."
- Natural accumulation of evidence: last year's posterior is this year's prior.

Practical.

- **Uncertainty in derived quantities is easy to get.** Callback to Lab 6: for the LD50, a Bayesian draws many (a, b) pairs from the posterior and computes $-a/b$ for each; the spread of those values is the posterior for the LD50. No profile, no reparameterization.
- **Hierarchical models.** Plots within sites within regions, animals measured repeatedly, many species sharing information: these models have many parameters, some describing variation among groups. They are awkward to fit by maximum likelihood and natural in a Bayesian framework (later in the semester).
- **Hidden quantities.** True occupancy when detection is imperfect (the frog from Lab 5), true infection status behind a noisy test, the true population size behind a count. These are just more unknowns in a Bayesian model.
- **Small samples.** No reliance on large-sample approximations like the Hessian interval, which goes negative for the Week 7 chytrid data (2 of 20 positive).
- **Prior information.** Previous studies, expert knowledge, or physical limits can be built in openly.

Costs. Priors must be chosen and defended; computation can be slow; MCMC output has to be checked for convergence.

6. Return to the big picture (~1 min)

- Same model, same likelihood. Bayesian inference multiplies by a prior and reads uncertainty off a posterior distribution.
- With one parameter and a conjugate prior, it's just adding counts. With many parameters, we need MCMC (Week 9).
- Hand out the Newton's method take-home (due Wednesday). Every MLE so far came from calculus, a grid, or `optim()` as a black box; Wednesday opens the box.

Discussion (~8 min)

Pick one or two. (The earlier "flat on p vs. flat on the log-odds" question is dropped; Week 7 Wednesday covered it.)

1. With the supplier prior, the posterior mean germination rate is 0.62; with a flat prior it's 0.74. A restoration manager says, "Keep the supplier's numbers out of my estimate." Is that reasonable? What would you report to them?
2. In Lab 6 you built a profile interval for the LD50. How would a Bayesian get an interval for the LD50, and which approach would you rather use for a model with ten parameters?
3. The Week 7 likelihood ratio interval (95%) for the pikas was about [1.8, 4.6]. A land manager asks, "So there's a 95% chance the true density is between 1.8 and 4.6?" Can you say yes? What could a Bayesian say instead?

Question 1 surfaces the "priors are cheating" worry. The manager has a point (the prior is worth 200 seeds to the data's 40, and it comes from other lots and an interested party), but the earlier lots aren't worthless either. Where to land: report both posteriors (a prior sensitivity analysis), perhaps with a down-weighted prior in between (the supplier's 60% scaled down to 12 of 20 seeds, Beta(13, 9)). Whether past lots are relevant to this one is an ecological question, not a statistical one. Question 2 connects today to Lab 6 and sets up MCMC. Question 3 lands on the credible vs. confidence distinction from 4(e).

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Full rationale and code: Part 1.

WOD ~8 min. Lecture ~32 min. Discussion ~8 min.

WOD debrief: tortoise analysts

- Shared likelihood peaks ~0.70 (21 of 30); read it off A (weak prior)
- Prior dependence: B > C > A (priors worth 50, 20, 4 vs. 30 tortoises)
- **B's narrow posterior = its prior, not more data**
- Manager: 0.62 to 0.84 from one dataset. Report the sensitivity; prior choice is ecological
- Bridge: supplier prior worth 200 seeds vs. 40 of data

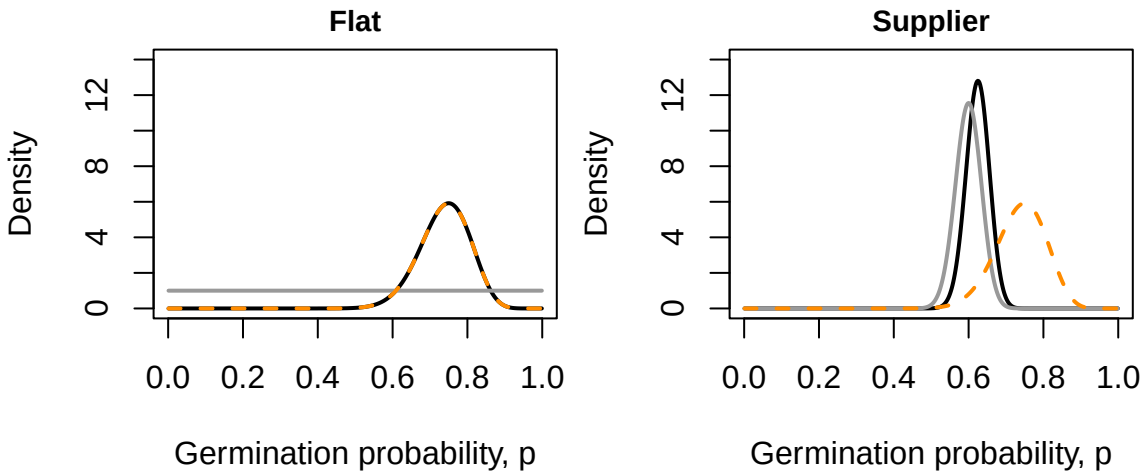
1. Big picture and recap (~3 min)

- Loop on board: THEORY → MODEL → DATA (fit) → COMPARE → repeat
- Posterior $\propto$ likelihood $\times$ prior
- Beta(α, β) + k of n → Beta($\alpha + k$, $\beta + n - k$)
- Today: how much does the prior matter?

2. Germination, two priors (~10 min)

Seeds: **30 of 40** germinate. Supplier: 120 of 200.

- Beta prior $\times$ binomial → Beta($\alpha + k$, $\beta + n - k$)
- Conjugate = add counts; $\alpha - 1$, $\beta - 1$ = prior successes, failures
- Flat: Beta(31, 11); mean 0.738, mode 0.75 = MLE
- Supplier: Beta(151, 91); mean 0.624
- 300 of 400 with supplier prior: mean 0.699
- Sequential updating = same answer


Flat
Density
0
4
8
12
0.0
0.2
0.4
0.6
0.8
1.0
Germination probability, p
Supplier
Density
0
4
8
12
0.0
0.2
0.4
0.6
0.8
1.0
Germination probability, p

3. Summaries (~4 min)

- Mean (usual), median, mode (= MLE if flat)
- Credible interval: quantiles or HPD
- Flat: [0.60, 0.86]; supplier: [0.56, 0.68]
- "95% probability p is in here" is allowed

4. Where people get stuck (~8 min)

- (a) Likelihood $\neq$ posterior (Week 5 WOD)
- (b) Flat isn't flat on every scale (callback only; covered Oct. 7)
- (c) Priors: report them; sensitivity analysis
- (d) Bayes $\neq$ MCMC
- (e) Credible $\neq$ confidence (pikas: [1.82, 4.61] vs. [1.91, 4.74])
- (f) Posterior is conditional on the model; still check and compare

5. Why so popular (~5 min)

- Direct probability statements
- Evidence accumulates: posterior $\rightarrow$ next prior
- Derived quantities: LD50 $= -a/b$ for each posterior draw (Lab 6)
- Hierarchical models (later)
- Hidden quantities: occupancy, true infection status
- Small samples: no Hessian approximation (chytrid)
- Costs: priors, computation, convergence

6. Wrap (~1 min)

- Conjugate = add counts; many parameters = MCMC (Week 9)
- **Hand out Newton take-home (due Wed.)**

Discussion (~8 min)

1. Supplier prior: mean 0.62; flat: 0.74. Manager: "keep the supplier's numbers out." Reasonable? What to report?
2. Bayesian interval for the LD50? Which approach for 10 parameters?
3. Week 7 pikas [1.8, 4.6]: "95% chance"? Can you say yes?